

2023 Ph H1 Q8

Section: Our Dynamic Universe

Topic: The Expanding Universe

Question:

Evidence supporting the existence of dark energy comes from:

- A. Estimations of the mass of galaxies
 - B. The darkness of the sky (Olbers' paradox)
 - C. Large numbers of galaxies showing redshift
 - D. The accelerating rate of expansion of the Universe
 - E. The abundance of hydrogen and helium in the Universe
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Answer:

D

Revision Tips:

- **Dark energy** is required to explain the **accelerating** expansion of the Universe
- **Redshift alone** supports expansion, not acceleration
- **Dark matter** is linked to galaxy mass
- **Olbers' paradox** explains why the sky is dark
- **Elemental abundance** is evidence for the **Big Bang**, not dark energy

